

Z-Score in Univariate Analysis

Z-Score in Data Science

Introduction

Z-score is an important statistical method used in Data Science and Univariate Analysis. It helps to measure how far a data value is from the mean of the dataset. The distance is measured using standard deviation. Z-score is mainly used for outlier detection, data standardization, and improving Machine Learning model performance.

In simple terms, Z-score tells whether a value is above the mean, below the mean, or exactly equal to the mean.

Formula of Z-Score

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- **X** = Data value
- **μ** = Mean of the dataset
- **σ** = Standard deviation

Meaning of Z-Score

Z-Score Value	Meaning
Z = 0	Value is equal to mean
Z > 0	Value is above mean
Z < 0	Value is below mean
Z = +2	2 standard deviations above mean
Z = -2	2 standard deviations below mean

Z-Score in Univariate Analysis

Importance of Z-Score

Z-score is very important in Data Science because it helps to understand the distribution of data and identify unusual values.

Main Uses of Z-Score

1. Outlier Detection

The major use of Z-score is identifying outliers.

If a data value is very far from the mean, it is called an outlier.

Generally:

$$|Z| > 3$$

Then the value is considered an outlier.

Outliers can affect:

- Mean
- Standard deviation
- Machine Learning model accuracy

So, identifying outliers is very important during data preprocessing.

2. Data Standardization

Datasets may contain different ranges of values.

Example:

- Age → 18 to 60
- Salary → 20,000 to 500,000

Large values dominate smaller values during model training.

Z-score converts all values on a common scale:

Z-Score in Univariate Analysis

- Mean becomes 0
- Standard deviation becomes 1

This process is called standardization.

3. Machine Learning Improvement

Many Machine Learning algorithms perform better after standardization.

Algorithms that require scaling:

- KNN
- SVM
- Logistic Regression
- PCA
- Neural Networks

Z-score helps these algorithms give better accuracy and faster performance.

Advantages of Z-Score

- Easy to calculate
- Useful for detecting outliers
- Helps in feature scaling
- Improves Machine Learning performance
- Works well for normally distributed data

Z-Score in Univariate Analysis

Limitations of Z-Score

- Sensitive to extreme outliers
- Works best only for normal distribution
- Not suitable for highly skewed data

For skewed data, the IQR method is preferred.

Z-Score in Univariate Analysis

Univariate Analysis means analyzing one variable at a time.

In univariate analysis, Z-score helps to:

- Understand data distribution
- Detect abnormal values
- Improve data quality
- Prepare data for Machine Learning

Example:

If we analyze only the “Salary” column, Z-score helps identify unusually high or low salary values.

Python Code for Z-Score

```
from scipy.stats import zscore
import pandas as pd
data = pd.DataFrame({
    'Marks':[45,50,52,48,47,300]
})
data["Z_score"] = zscore(data["Marks"])

print(data)
```

Z-Score in Univariate Analysis

Interview Questions

What is Z-score?

Z-score measures how many standard deviations a data point is away from the mean.

Why is Z-score used?

It is used for:

- Outlier detection
- Standardization
- Improving Machine Learning models

What is considered an outlier?

If:

$$|Z| > 3$$

the value is generally considered an outlier.

Which algorithms need standardization?

- KNN
- SVM
- Logistic Regression
- PCA

These algorithms are sensitive to feature scaling.

Conclusion

Z-score is an essential concept in Data Science and Statistics. It is widely used for outlier detection, feature scaling, and data preprocessing. In Univariate Analysis, Z-score helps to understand the behavior of a single variable and identify abnormal values. It is one of the most asked topics in Data Analyst and Data Scientist interviews.